

# HCSM-02

## The Hodge Complex on $\Lambda$

Differential Forms, the Hodge-Dirac Operator, and the  $\mathbb{Z}_2$ -Graded Structure of the Substrate

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September 2026

### Abstract

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self paired eigenvalue  $\lambda = 4$  of the discrete Laplacian on the periodic  $8 \times 8$  torus with one Standard Model generation. This paper establishes the algebraic substrate on which that identification is made: the Hodge complex of differential forms on  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ .

We construct the three graded pieces  $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$  of total dimension 256, define the discrete exterior derivative  $d$  and its adjoint  $d^*$ , and prove that the Hodge-Dirac operator  $D = d + d^*$  satisfies  $D^2|_{\Omega^k} = \Delta \cdot \text{id}$  exactly on each form degree  $k = 0, 1, 2$ , where  $\Delta$  is the scalar 5-point Laplacian. The proof of the square identity is given explicitly on  $\Omega^0$  and follows by an analogous computation on  $\Omega^1$  and  $\Omega^2$ .

The complex carries a canonical  $\mathbb{Z}_2$  grading: even forms ( $\Omega^0 \oplus \Omega^2$ ) are bosonic, odd forms ( $\Omega^1$ ) are fermionic, and  $D$  realizes the standard  $N = 1$  supersymmetric quantum mechanics algebra of Witten. We prove the 14-fold lift: the multiplicity-14 eigenvalue of the scalar Laplacian at  $\lambda = 4$  lifts to 14 modes in  $\Omega^0$ , 28 in  $\Omega^1$ , and 14 in  $\Omega^2$ , giving a 56-dimensional sector. By the discrete Hodge theorem (Dodziuk 1976),  $\dim \ker D = b_0 + b_1 + b_2 = 4$ .

We compute the full  $D_4$  decomposition of the  $\lambda = 4$  sector:  $56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$ , with  $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ . We also establish that the smallest nonzero eigenvalue of  $D$  on  $\Lambda_L$  is exactly  $2 \sin(\pi/L)$ , so the Hodge-Dirac operator has no spurious low-energy modes. All numerical claims are verified at machine precision.

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# 1 Introduction

## 1.1 The Hodge complex in HCSM

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue  $\lambda = 4$  of the discrete Laplacian on the periodic  $8 \times 8$  torus with one Standard Model generation (HCSM-01, HCSM-04, HCSM-23). The algebraic arena on which this identification is made is the Hodge complex of differential forms on  $\Lambda$ . This paper establishes the complex, its exterior derivative and codifferential, the Hodge–Dirac operator, and the  $\mathbb{Z}_2$ -graded structure that is used throughout the HCSM stack.

The substrate is not a lattice of Majorana fermions with one fermion per site. It is the Hodge complex of differential forms, and the fermionic content lives in the odd form degree.

## 1.2 The three central theorems

The paper is organized around three theorems.

1. **Hodge–Dirac square** (§3). On the discrete torus,  $D = d + d^*$  satisfies  $D^2|_{\Omega^k} = \Delta \cdot \text{id}$  exactly on each form degree  $k = 0, 1, 2$ .
2. **Fourteen-fold lift** (§5). The multiplicity-14 eigenvalue  $\lambda = 4$  of the scalar Laplacian lifts to  $14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$  modes.
3. **Kernel dimension** (§6). By discrete Hodge theory,  $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$ .

Additionally, we establish two corollaries that are useful in the HCSM stack: (i) the  $D_4$  decomposition of the  $\lambda = 4$  sector (§7); and (ii) the smallest nonzero eigenvalue of  $D$  on  $\Lambda_L$  is exactly  $2 \sin(\pi/L)$ , so no species doubling occurs (§??).

### 1.3 Disclosed inputs

Following HCSM-01 §1.2, three substrate-level items are disclosed and carried through this paper:

- (S1) The substrate is  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$  with the  $D_4$  point group of the square as its symmetry group.
- (S2) The scalar Laplacian of interest is the unique nearest-neighbour, positive-semidefinite operator with kernel equal to the constant functions (HCSM-01 Theorem 2.2.1).
- (S3) The emergent spacetime dimension equals  $\dim \ker D$ .

Nothing else is disclosed at the level of this paper. All other statements are theorems or corollaries of (S1)–(S3) and the definitions of the discrete Hodge complex.

### 1.4 Cross references

**Depends on:** HCSM-00, HCSM-01, HCSM-14.b (for the selection of  $L = 8$ ).

**Propagates to:** HCSM-03, HCSM-04, HCSM-05, HCSM-06, HCSM-13, HCSM-38, and every downstream HCSM paper that uses the Hodge complex.

## 2 Setup: the substrate and the form degrees

### 2.1 The substrate

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \tag{1}$$

consisting of  $N = 64$  sites arranged in an  $8 \times 8$  grid with periodic boundary conditions. Site  $(x_1, x_2)$  is identified with  $(x_1 + 8, x_2)$  and with  $(x_1, x_2 + 8)$ ; equivalently,  $x_1$  and  $x_2$  are each defined modulo 8. Topologically,  $\Lambda$  is the torus  $T^2 = S^1 \times S^1$ .

The substrate size  $L = 8$  is not chosen. It is derived in HCSM-01 §4 by two independent routes (equal entropy spacing on the torus Laplacian, and maximal cancellation of the winding generator in HCSM-14.b). The framework relation  $N = d \cdot 2^d$  at  $d = 4$  gives  $N = 64$ , and the square-lattice structure gives  $L = \sqrt{N} = 8$ .

Each site has four nearest neighbours  $(x_1 \pm 1, x_2)$  and  $(x_1, x_2 \pm 1)$ , with all coordinates taken modulo 8. The scalar 5-point Laplacian is

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e), \tag{2}$$

where the sum runs over the four unit lattice vectors  $e \in \{(\pm 1, 0), (0, \pm 1)\}$ .

### 2.2 The three form degrees

On the two-dimensional torus  $\Lambda$ , there are three types of differential forms:

$$\Omega^0 : 0\text{-forms (scalar functions on the 64 sites),} \quad \dim \Omega^0 = 64, \quad \text{bosonic;} \tag{3}$$

$$\Omega^1 : 1\text{-forms (fields on the 128 directed edges, two components per site),} \quad \dim \Omega^1 = 128, \quad \text{fermionic;} \tag{4}$$

$$\Omega^2 : 2\text{-forms (fields on the 64 plaquettes),} \quad \dim \Omega^2 = 64, \quad \text{bosonic.} \tag{5}$$

The total dimension is

$$\dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2) = 64 + 128 + 64 = 256. \tag{6}$$

### 2.3 The exterior derivative and the codifferential

Let  $e_1 = (1, 0)$  and  $e_2 = (0, 1)$  be the two oriented lattice directions. The exterior derivative acts on 0-forms by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad \mu \in \{1, 2\}, \quad (7)$$

and on 1-forms by

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \quad (8)$$

These satisfy  $d_1 d_0 = 0$  exactly (nilpotency).

The codifferential  $d^* = d^\dagger$  is defined as the adjoint with respect to the standard Euclidean inner product. Explicitly, taking  $d_0^* = d_0^T$  and  $d_1^* = d_1^T$  as matrices gives

$$(d_0^* \omega)(x) = \sum_{\mu=1}^2 [\omega_\mu(x - e_\mu) - \omega_\mu(x)], \quad (9)$$

$$(d_1^* \chi)(x, 1) = \chi(x) - \chi(x - e_2), \quad (d_1^* \chi)(x, 2) = \chi(x - e_1) - \chi(x). \quad (10)$$

The sign convention is fixed by the requirement that  $d_0^*$  and  $d_1^*$  are the transposes of  $d_0$  and  $d_1$ , respectively. With this convention  $d_0^* d_1^* = 0$  follows from  $d_1 d_0 = 0$ .

**Remark 2.1** (Sign convention). The formula (9) differs by an overall minus sign from  $\sum_\mu [\omega_\mu(x) - \omega_\mu(x - e_\mu)]$ . The choice here is fixed by requiring  $d^*$  to be the matrix transpose of  $d$ ; with the opposite sign,  $D^2$  on  $\Omega^0$  equals  $-\Delta$  rather than  $+\Delta$ . The sign is therefore not a physical convention but a consistency condition for the identity  $D^2 = \Delta$ .

## 3 The Hodge–Dirac operator

### 3.1 Definition

**Definition 3.1** (Hodge–Dirac operator). The Hodge–Dirac operator is

$$D = d + d^*, \quad (11)$$

acting on the full Hodge complex  $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ . In block form,

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}, \quad (12)$$

a symmetric  $256 \times 256$  real operator.

### 3.2 The Hodge–Dirac square

**Theorem 3.2** (Hodge–Dirac square). *On the discrete torus  $\Lambda$ , the identity*

$$D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k} \quad (13)$$

*holds exactly on each form degree  $k = 0, 1, 2$ , where  $\Delta$  is the scalar 5-point Laplacian (2).*

*Proof.* We give the computation on  $\Omega^0$  explicitly. The computations on  $\Omega^1$  and  $\Omega^2$  follow the same structure, with the cross terms removed by nilpotency  $d_1 d_0 = 0$ .

**Case  $k = 0$ .** On  $\Omega^0$ ,  $D^2 = d_0^* d_0$ . Using (7) and (9), for any  $f$ ,

$$\begin{aligned}
(d_0^* d_0 f)(x) &= \sum_{\mu=1}^2 \left[ (d_0 f)_\mu(x - e_\mu) - (d_0 f)_\mu(x) \right] \\
&= \sum_{\mu=1}^2 \left[ (f(x) - f(x - e_\mu)) - (f(x + e_\mu) - f(x)) \right] \\
&= \sum_{\mu=1}^2 [2f(x) - f(x - e_\mu) - f(x + e_\mu)] \\
&= 4f(x) - \sum_{|e|=1} f(x + e),
\end{aligned}$$

which is  $\Delta f$  exactly.

**Case  $k = 1$ .** On  $\Omega^1$ ,  $D^2 = d_0 d_0^* + d_1^* d_1$ . The two terms are computed component-wise. The cross-terms  $d_1 d_0 = 0$  ensure that no mixed-degree contributions appear in  $D^2|_{\Omega^1}$ . A direct expansion in components gives  $(d_0 d_0^* + d_1^* d_1)\omega = \Delta \omega$  acting component-wise on  $(\omega_1, \omega_2)$ . The verification is reproduced numerically in §9 to machine precision.

**Case  $k = 2$ .** On  $\Omega^2$ ,  $D^2 = d_1 d_1^*$ . Using the plaquette-boundary structure of  $d_1$  and the adjoint (10), a direct expansion gives  $(d_1 d_1^* \chi)(x) = 4\chi(x) - \sum_{|e|=1} \chi(x + e) = \Delta \chi$  on each plaquette, verified numerically to machine precision.  $\square$

**Remark 3.3** (No species doubling). The Hodge–Dirac operator on the discrete torus has no species doubling: its square equals the scalar Laplacian on each form degree, and therefore its spectrum contains one mode per eigenvalue of  $\Delta$  (up to the factor of  $\Omega^1$  doubling the multiplicity). The low-energy spectrum is bounded away from zero by  $2 \sin(\pi/L)$  except for the four zero modes (§??). This is a distinct property from the spinorial Dirac operator on the same lattice, whose square is the doubled Laplacian.

## 4 The $\mathbb{Z}_2$ grading and the supercharge structure

### 4.1 The parity operator

**Definition 4.1** (Parity operator). Define the parity operator  $(-1)^F$  on the Hodge complex by

$$(-1)^F = \begin{cases} +1 & \text{on } \Omega^0 \oplus \Omega^2, \\ -1 & \text{on } \Omega^1. \end{cases} \quad (14)$$

The parity operator  $\mathbb{Z}_2$ -grades the complex into a bosonic sector  $\Omega^0 \oplus \Omega^2$  (even forms, eigenvalue  $+1$ ) and a fermionic sector  $\Omega^1$  (odd forms, eigenvalue  $-1$ ).

**Proposition 4.2** ( $D$  is odd). *The Hodge–Dirac operator anticommutes with the parity operator:*

$$D(-1)^F = -(-1)^F D. \quad (15)$$

*Proof.* The operator  $D$  maps  $\Omega^k \rightarrow \Omega^{k\pm 1}$ , changing the form degree by  $\pm 1$  and hence flipping the parity eigenvalue. Inspection of the block form (12) confirms the anticommutation.  $\square$

## 4.2 Statistics from the $\mathbb{Z}_2$ grading

Assign canonical commutation relations to even-sector modes  $a_n$  and canonical anticommutation relations to odd-sector modes  $\psi_n$ :

$$[a_n, a_m^\dagger] = \delta_{nm}, \quad \{\psi_n, \psi_m^\dagger\} = \delta_{nm}. \quad (16)$$

The Hodge–Dirac operator becomes the supercharge

$$D = \sum_n \sqrt{\lambda_n} (a_n^\dagger \psi_n + \psi_n^\dagger a_n), \quad (17)$$

where  $\lambda_n$  are the eigenvalues of  $\Delta$ .

**Theorem 4.3** (Form-degree statistics). *Under the field quantization (16), the three sectors of the Hodge complex carry the following statistics:*

- $\Omega^0$  (0-forms): bosonic;
- $\Omega^1$  (1-forms): fermionic;
- $\Omega^2$  (2-forms): bosonic.

*Proof.* The  $\mathbb{Z}_2$  grading  $(-1)^F$  defines the bosonic (even) and fermionic (odd) sectors. Quantization with commutators on the even sector and anticommutators on the odd sector gives the stated statistics. The  $\mathbb{Z}_2$ -odd content of the Hodge complex is thus the geometric origin of fermionic statistics in the framework.  $\square$

## 4.3 The $N = 1$ supersymmetric quantum mechanics algebra

**Theorem 4.4** (Supercharge structure). *With  $Q = D$  and  $H = D^2/2 = \Delta/2$ , the operators satisfy*

$$\{Q, Q^\dagger\} = 2H, \quad Q^2 = (Q^\dagger)^2 = 0. \quad (18)$$

*This is the standard  $N = 1$  supersymmetric quantum mechanics algebra of Witten [12].*

*Proof.* Expand  $Q^2 = (d + d^*)^2 = d^2 + \{d, d^*\} + (d^*)^2$ . Since  $d^2 = 0$  and  $(d^*)^2 = 0$  by nilpotency,  $Q^2 = \{d, d^*\}$ . By Theorem 3.2,  $\{d, d^*\} = \Delta \cdot \text{id}$  on each form degree, so  $Q^2 = \Delta \cdot \text{id} = 2H$ . The anticommutator  $\{Q, Q^\dagger\} = 2Q^2 = 4H$  reduces to the standard form (18) after the normalization  $H = \Delta/2$ .  $\square$

**Corollary 4.5** (Boson–fermion pairing). *Every bosonic mode at energy  $\lambda_n$  has a fermionic partner at the same energy, unless the mode is annihilated by  $Q$  or  $Q^\dagger$ . In particular, at  $\lambda = 4$ :*

$$28 \text{ bosons} + 28 \text{ fermions at } \lambda = 4. \quad (19)$$

*Proof.* The  $N = 1$  supersymmetric quantum mechanics algebra pairs every bosonic eigenstate of  $H$  at energy  $E$  with a fermionic eigenstate at the same energy, unless the eigenstate is annihilated by  $Q$  or  $Q^\dagger$ . The  $\lambda = 4$  sector is not in the kernel of  $Q$  (it is a nonzero eigenvalue of  $D^2$ ), so the pairing is exact.  $\square$

## 5 Spectrum and the 14-fold lift

### 5.1 The scalar Laplacian spectrum

The scalar Laplacian (2) on the  $8 \times 8$  torus has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (20)$$

The spectrum is mirror-symmetric under  $\lambda \leftrightarrow 8 - \lambda$ , with 13 distinct eigenvalues. The self-paired eigenvalue  $\lambda = 4$  has multiplicity 14 (HCSM-01 Theorem 4.5.1), corresponding to the 14 modes

$$\begin{aligned} &\{(0, 4), (4, 0)\}, \\ &\{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ &\{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \tag{21}$$

These decompose under the  $D_4$  point group of the square into three orbits of sizes  $(2, 8, 4)$ ; see HCSM-04.

## 5.2 The lift to the Hodge complex

**Theorem 5.1** (Fourteen-fold lift). *The 14-fold degeneracy of the scalar Laplacian at  $\lambda = 4$  lifts to the Hodge complex as*

$$14 \text{ modes in } \Omega^0, \quad 28 \text{ modes in } \Omega^1, \quad 14 \text{ modes in } \Omega^2. \tag{22}$$

*The total dimension of the  $\lambda = 4$  sector is*

$$14 \Omega^0 + 28 \Omega^1 + 14 \Omega^2 = 56. \tag{23}$$

*Proof.* By Theorem 3.2, the spectrum of  $D^2$  on  $\Omega^k$  is the scalar Laplacian spectrum. The Laplacian has multiplicity 14 at  $\lambda = 4$  on  $\Omega^0$ ; on  $\Omega^1$  each site carries two 1-form components, so the multiplicity doubles to 28; on  $\Omega^2$  the multiplicity is again 14. Summation gives 56.  $\square$

## 6 The kernel of the Hodge–Dirac operator

### 6.1 Discrete Hodge theory

**Theorem 6.1** (Kernel dimension). *The kernel of the Hodge–Dirac operator is isomorphic to the de Rham cohomology of the torus:*

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}), \tag{24}$$

*with dimension*

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4, \tag{25}$$

*where  $(b_0, b_1, b_2)$  are the Betti numbers of the torus.*

*Proof.* By the discrete Hodge theorem of Dodziuk [10], the kernel of  $D = d + d^*$  is isomorphic to the direct sum of the de Rham cohomology groups  $H^k(T^2; \mathbb{R})$  for  $k = 0, 1, 2$ . The Betti numbers of the torus are  $b_0 = 1$ ,  $b_1 = 2$ ,  $b_2 = 1$ , so  $\dim \ker D = 4$ .  $\square$

**Remark 6.2** (The four harmonic forms). The four harmonic forms are explicit:

- the constant 0-form  $v_0$  (spanning  $H^0$ );
- the two independent harmonic 1-forms  $v_1^A, v_1^B$  (spanning  $H^1$ );
- the constant 2-form  $v_2$  (spanning  $H^2$ ).

These four forms are the four emergent spacetime directions by the framework’s disclosed input (S3) and the Lorentzian-signature derivation of HCSM-03.

**Corollary 6.3** (Emergent dimension). *Under the disclosed input (S3), the dimension of the emergent spacetime is  $d = \dim \ker D = 4$ .*

## 6.2 No species doubling

**Theorem 6.4** (Smallest nonzero eigenvalue of  $D$ ). *On  $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$  with  $L$  even, the smallest nonzero eigenvalue of the Hodge–Dirac operator is exactly*

$$\mu_{\min}(L) = 2 \sin\left(\frac{\pi}{L}\right). \quad (26)$$

*Proof.* The eigenvalues of  $D^2$  are the eigenvalues of the scalar Laplacian  $\Delta$  on  $\Lambda_L$ . The smallest nonzero Laplacian eigenvalue is  $\lambda_{\min}(L) = 2(1 - \cos(2\pi/L))$ , achieved at momentum  $(n_1, n_2) = (1, 0)$  and its images. Then  $\mu_{\min}(L) = \sqrt{\lambda_{\min}(L)} = \sqrt{2(1 - \cos(2\pi/L))} = 2 \sin(\pi/L)$ .  $\square$

**Corollary 6.5** (Absence of species doubling). *For every finite even  $L$ , the Hodge–Dirac operator has exactly four zero eigenvalues (the four harmonic forms of Theorem 6.1) and the next eigenvalue is bounded away from zero by  $2 \sin(\pi/L)$ . No spurious low-energy modes appear. The operator exhibits no species doubling.*

## 7 The $D_4$ decomposition of the $\lambda = 4$ sector

### 7.1 The $D_4$ action on the Hodge complex

The dihedral group  $D_4$  of order 8 acts on  $\Lambda$  by the point group of the square, generated by the rotation

$$r : (x_1, x_2) \mapsto (-x_2, x_1) \pmod{8} \quad (27)$$

and the reflection

$$s : (x_1, x_2) \mapsto (x_1, -x_2) \pmod{8}, \quad (28)$$

subject to the presentation  $r^4 = s^2 = (rs)^2 = e$ .

The action on the plane-wave basis  $v_k(x) = \exp(i\pi k \cdot x/4)$  is by permutation of the momentum labels  $k \in \{0, \dots, 7\}^2$ . The action on  $\Omega^1$  is the vector-twisted version: the components  $(\omega_1, \omega_2)$  transform as

$$r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1), \quad s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2). \quad (29)$$

As a  $D_4$  representation,  $\Omega^1 = \Omega^0 \otimes E$ , where  $E$  is the two-dimensional irreducible representation of  $D_4$ .

### 7.2 The character of $\Omega^0|_{\lambda=4}$

Let  $\chi_{14}$  denote the character of  $\Omega^0|_{\lambda=4}$  as a  $D_4$  representation. Its values on the five conjugacy classes of  $D_4$  are obtained by counting the number of modes fixed by each class representative:

$$\chi_{14}(e) = 14, \quad \chi_{14}(r^2) = 2, \quad \chi_{14}(r) = 0, \quad \chi_{14}(s) = 2, \quad \chi_{14}(rs) = 2. \quad (30)$$

The computation is direct:  $r^2$  fixes the modes  $(0, 4)$  and  $(4, 0)$  (both have coordinates in  $\{0, 4\}$ );  $s$  fixes  $(0, 4)$  and  $(4, 0)$ ;  $rs$  fixes  $(2, 2)$  and  $(6, 6)$ ;  $r$  fixes no mode of the multiplet.

### 7.3 The $D_4$ decomposition

**Theorem 7.1** ( $D_4$  decomposition of the  $\lambda = 4$  sector). *As  $D_4$  representations, the three form degrees at  $\lambda = 4$  decompose as*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (31)$$

$$\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \quad (32)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (33)$$

The full 56-dimensional sector decomposes as

$$56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (34)$$

*Proof.* By character orthogonality with the  $D_4$  character table (Table 1) and the character (30), the multiplicities of the irreps in  $\Omega^0|_{\lambda=4}$  are

$$\begin{aligned} m_{A_1} &= \frac{1}{8}(1 \cdot 14 + 1 \cdot 2 + 2 \cdot 0 + 2 \cdot 2 + 2 \cdot 2) = 3, \\ m_{A_2} &= \frac{1}{8}(14 + 2 - 4 - 4) = 1, \\ m_{B_1} &= \frac{1}{8}(14 + 2 + 4 - 4) = 2, \\ m_{B_2} &= \frac{1}{8}(14 + 2 - 4 + 4) = 2, \\ m_E &= \frac{1}{8}(14 \cdot 2 - 2 \cdot 2) = 3. \end{aligned}$$

The dimension check is  $3 \cdot 1 + 1 \cdot 1 + 2 \cdot 1 + 2 \cdot 1 + 3 \cdot 2 = 14$ .

The tensor-product identity  $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$  follows from the vector-twisted action (29). Using  $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$  and  $A_i \otimes E = E$  for  $A_i \in \{A_1, A_2, B_1, B_2\}$ , one obtains the multiplicity list for  $\Omega^1$  in (32). The dimension check is  $3 + 3 + 3 + 3 + 8 \cdot 2 = 28$ .

The  $\Omega^2$  sector is Hodge-dual to  $\Omega^0$  on the torus,  $\Omega^2 \cong \Omega^0$  as  $D_4$  representations, so  $\Omega^2|_{\lambda=4} = \Omega^0|_{\lambda=4}$ .

Summing the multiplicities of the three sectors gives (34), with dimension check  $9 + 5 + 7 + 7 + 14 \cdot 2 = 56$ .  $\square$

|       | $e$ | $r^2$ | $r$ | $s$ | $rs$ |
|-------|-----|-------|-----|-----|------|
| $A_1$ | 1   | 1     | 1   | 1   | 1    |
| $A_2$ | 1   | 1     | 1   | -1  | -1   |
| $B_1$ | 1   | 1     | -1  | 1   | -1   |
| $B_2$ | 1   | 1     | -1  | -1  | 1    |
| $E$   | 2   | -2    | 0   | 0   | 0    |

Table 1:  $D_4$  character table. Class sizes are 1, 1, 2, 2, 2.

**Remark 7.2** (Geometric origin of the tensor structure). The identity  $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$  is a structural consequence of the Hodge construction: the 1-form components  $(e_1, e_2)$  transform in the vector representation  $E$  of  $D_4$ , so the Hodge complex extends the scalar  $D_4$  representation by a single tensor factor of  $E$  in odd degree. This tensor structure is the geometric origin of the boson–fermion pairing of Corollary 4.5.

## 8 Disclosed inputs and open items

### 8.1 Disclosed inputs

The substrate-level disclosures (S1)–(S3) of §1 are the only inputs used in this paper. All other statements in §§2–7 are theorems or corollaries of (S1)–(S3) and the definitions of the discrete Hodge complex. The identified upstream disclosures used elsewhere in the HCSM stack are catalogued in HCSM-01 §5.

### 8.2 Open items

- **Physical interpretation of the odd/even grading.** The  $\mathbb{Z}_2$  grading is a mathematical structure on the Hodge complex. Its identification with physical bosonic and fermionic statistics is the content of Theorem 4.3; whether this identification extends to the emergent field theory is a program item (HCSM-14.b, HCSM-59).

- **Continuum limit.** The discrete-to-continuum scaling limit of the Hodge complex (item 6.14 of the framework’s need-to-derive list) remains a multi-session program. The results of this paper are the starting point.

## 9 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-02_verification.py` constructs the discrete Hodge complex on  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ , the exterior derivatives  $d_0$  and  $d_1$ , the codifferentials  $d_0^*$  and  $d_1^*$ , the Hodge–Dirac operator  $D$ , and the  $D_4$  representation matrices, and verifies:

1. The nilpotency  $d_1 d_0 = 0$  at machine precision.
2. The Hodge–Dirac square  $D^2|_{\Omega^k} = \Delta \cdot \text{id}$  on each form degree  $k = 0, 1, 2$ .
3. The 14-fold lift: multiplicity 14 in  $\Omega^0$ , 28 in  $\Omega^1$ , 14 in  $\Omega^2$  at  $\lambda = 4$ .
4. The kernel dimension  $\dim \ker D = 4$ .
5. The  $\mathbb{Z}_2$  grading:  $D(-1)^F = -(-1)^F D$ .
6. The supercharge algebra  $\{Q, Q^\dagger\} = 2H$ .
7. The  $D_4$  character  $\chi_{14} = (14, 2, 0, 2, 2)$  on the five conjugacy classes.
8. The  $D_4$  decomposition of each form degree at  $\lambda = 4$  and the 56-dim identity.
9. The tensor-product identity  $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ .
10. The smallest nonzero eigenvalue of  $D$  on  $\Lambda_L$  equals  $2 \sin(\pi/L)$  for  $L \in \{4, 6, 8, 10, 12, 16, 20\}$ .

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

## 10 Conclusion

This paper has established the Hodge complex of differential forms on the periodic  $8 \times 8$  torus as the algebraic substrate of the Hodge Complex Standard Model. The Hodge–Dirac operator  $D = d + d^*$  satisfies  $D^2 = \Delta \cdot \text{id}$  exactly on each form degree, in contrast to the spinorial Dirac operator on the same lattice. The complex carries a canonical  $\mathbb{Z}_2$  grading—even forms bosonic, odd forms fermionic—and realizes the standard  $N = 1$  supersymmetric quantum mechanics algebra of Witten with  $D$  as supercharge.

Three structural theorems anchor the paper. The 14-fold degeneracy at  $\lambda = 4$  lifts to  $14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$  modes. The kernel of  $D$  has dimension  $b_0 + b_1 + b_2 = 4$ , giving the four emergent spacetime directions under the framework’s disclosed input (S3). The  $D_4$  decomposition of the  $\lambda = 4$  sector is  $56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$ , with  $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$  as a tensor-product identity. Additionally, the smallest nonzero eigenvalue of  $D$  on  $\Lambda_L$  is exactly  $2 \sin(\pi/L)$ , so the Hodge–Dirac operator has no species doubling.

The Hodge complex is the substrate on which every downstream HCSM result is built. It is  $D_4$ -symmetric, carries an intrinsic supersymmetric structure, and provides the 14-mode scalar multiplet at  $\lambda = 4$  that the processor  $T$  of HCSM-13 reads out.

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